

Lab 2: Human Subject Research

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1 Student Outcomes

By the end of this lab, students will be able to:

1. Develop and upload Arduino programs that use digital and analog input/output functions such as `digitalWrite()`, `digitalRead()`, `analogWrite()`, and `analogRead()`.
2. Build and test basic electronic circuits using breadboards, LEDs, switches, resistors, and potentiometers while interfacing them with an Arduino UNO.
3. Design and implement a simple embedded hardware system that integrates sensors, actuators, and control logic to satisfy specified functional requirements.

2 Equipment list

1. CITI website <https://about.citiprogram.org/>
2. UIC IRB website <https://research.uic.edu/compliance/human-subjects-irbs-2/>

3 Assignment: Human Research Refresher Course

Research involving human participants must be conducted ethically and responsibly. The purpose of this training is to introduce students to the principles of human-subjects research, participant safety, informed consent, privacy, confidentiality, and responsible experimental design.

1. Go to the CITI website <https://about.citiprogram.org/>. Click “Log In”, then click “Log In Through My Organization”, search for “University of Illinois Chicago,” then use your UIC ID to log in.
2. Complete the course “Group 1. HSP, Biomedical Research Investigators and Key Personnel”.
3. After completing the course, share the certificate with the teaching assistant and instructor.

4 Assignment: UIC IRB

This module is intended to help researchers conduct studies ethically and in compliance with university and federal requirements.

4.1 Readings

1. Watch the Office for Human Research Protections (OHRP) video on Institutional Review Boards (IRBs)
<https://www.youtube.com/watch?v=U8fme1boEbE>
2. Read the UIC Human Subjects Research and IRB overview website
<https://research.uic.edu/compliance/human-subjects-irbs-2/>
3. Read the UIC IRB Templates page to become familiar with consent forms, assent forms, and study documentation
<https://research.uic.edu/compliance/human-subjects-irbs-2/investigator/uic-research-irb-templates/>
4. Read the U.S. Department of Health and Human Services page on the Belmont Report ethical principles
<https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html>

4.2 Questions

1. What is the primary purpose of an Institutional Review Board (IRB)?
 - (a) To help researchers publish papers faster
 - (b) To ensure research participants are protected
 - (c) To increase research funding
 - (d) To evaluate student grades
2. Which of the following studies would most likely require IRB review?
 - (a) Testing a robot only in simulation
 - (b) Measuring battery voltage of a device
 - (c) Collecting walking data from human participants
 - (d) Calibrating a sensor on a workbench
3. What does informed consent mean?
 - (a) Participants are forced to complete the study
 - (b) Participants are paid before the experiment
 - (c) Participants understand the study and voluntarily agree to participate
 - (d) Participants must answer every question
4. Which of the following is an example of confidential information?
 - (a) Anonymous random numbers

- (b) Public weather data
 - (c) A participant's medical information
 - (d) The dimensions of a robot
5. According to ethical research practices, participants should:
- (a) Be allowed to stop participation at any time
 - (b) Be required to complete all experiments
 - (c) Never ask questions during the study
 - (d) Share their personal data publicly

Answers

- 1. (b) To ensure research participants are protected
- 2. (c) Collecting walking data from human participants
- 3. (c) Participants understand the study and voluntarily agree to participate
- 4. (c) A participant's medical information
- 5. (a) Be allowed to stop participation at any time